

Darell Barnes

(803) 290-6833 | darellbarnes450@gmail.com | LinkedIn | Portfolio | Github

Education

Voorhees University

Expected: May 2029

B.S. Computer Science, W. Franklin Evans Honors College

- GPA: 4.0
- Relevant Courses: Data Structures & Algorithms, Introduction to CS, Programming I&II

Leadership & Experience

Research Internship — CERPI, Voorhees University (Upcoming)

- Selected as an undergraduate research intern for a faculty-led project funded by a SCICU research grant on AI-guided bacteriophage design.
- Supporting computational evaluation of AI-generated bacteriophage sequences, analyzing biological validity, stability, and feasibility before experimental validation.
- Contributing to research aimed at assessing whether AI-assisted biological design can improve efficiency in developing phage-based therapies for antibiotic resistance.

VU CTRL Tech Club President

Sep 2025 – Present

- Directed student engineering initiatives focused on applied software development and project-based technical execution.
- Built and deployed an internal inventory management system to track and manage club resources and executive operations.
- Structured and led hands-on development sessions, guiding members through building functional software projects from design to implementation.

Mastercard X AUC Data Science Initiative – Competitor

Oct 2025 – Nov 2025

- Led a 4-person team in a forecasting competition focused on identifying economic growth opportunities across geographic tracts.
- Developed an end-to-end data pipeline in Python covering data preprocessing, feature engineering, regression modeling, and evaluation at the tract-level granularity.
- Presented model-driven findings and recommended optimal target tracts to competition organizers.

Projects

Barnes DB (Ongoing) | C, Systems Programming

June 2025 – Present

- Built a lightweight database engine to simulate core database operations using a custom query language and file-based storage.
- Implemented a token-based parser supporting INSERT, SELECT, UPDATE, and DELETE operations.
- Designed in-memory data structures with manual memory management, input validation, and error handling for low-level control.

AI IdeaBank | Python, Cloudflare Workers, Durable Objects, LLM

March 2026 – April 2026

- Built an AI-powered idea management system to help users capture, structure, and refine unorganized thoughts into persistent, retrievable records.
- Implemented an LLM-driven tool-calling workflow that converts natural language into structured actions for creating, updating, and deleting ideas.
- Designed persistent state management using Cloudflare Durable Objects to maintain chat history and user-specific data across sessions.

AuthBarn (PyPI Package) | Python, MySQL, Hashing

January 2025 – May 2025

- Developed a reusable authentication and authorization library to simplify secure user management for Python applications.
- Implemented password hashing, session handling, and role-based access control using object-oriented design principles.
- Packaged and published the library to PyPI, enabling installation via `pip install authbarn`.

Technical Skills

Languages: Python, C, JavaScript

Backend / Frameworks: Flask, FastAPI, HTMX

Databases: SQLite, MySQL, PostgreSQL

Core Concepts: REST APIs, Authentication Systems, Role-Based Access Control, Data Modeling

Tools: Git, GitHub, Linux, VS Code